COMP810: Data Warehouse and Big Data
Data Warehousing Project

(Weight: 50%)

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Project Overview

This assessment is designed to assess the following learning outcomes:

1. Critically reflect on in-demand data warehouse designs, their role in a decision-making process, advantages and potential problem areas.
2. Critically appraise data modelling options to design data warehouse schemas using contemporary patterns, including but not limited to Star and Snowflake.
3. Understand and demonstrate knowledge of advanced SQL techniques to access, aggregate and analyse data from big relational databases.

In a nutshell, it consists of two sections. In the first section, you need to design and implement a data warehouse. In the second section, you will write some queries to answer some specific questions based on a given schema.

1.1 Group Work

This is a Group Assignment (two members max). Liaise with your classmates and create your team of two people. Notes:

(a) Students can choose to work individually; there will be NO penalty. In other words, this Project will be marked up to the same standards and marking criteria, regardless of whether you work alone or in a group of two students.

(b) **Please form a group on Canvas by yourself by 2 Oct.** Alternatively, email name/s and student/s ID of your team (one or two students) to Ryan (ryan.ip@aut.ac.nz). You will be assumed to be working individually if you did not sign up or indicate your group.

Section A - Data Warehouse Schema: Design and Implementation (42 marks)

A case study for an Auckland-based business 'PhoneFix', our client, is given below. You have been asked:

(a) to choose a data warehouse design for the client, with explanations provided and,

(b) to draw a diagram to represent the logical schema you propose. You will then create the resulting data warehouse tables in Oracle.

A.1 Business Overview

'PhoneFix' sells mobile phone accessories and repairs mobile phones, with more than ten stores located within Auckland. Each of the 'PhoneFix' stores is independently run by a different owner (similar to a co-operative). As such, each owner manages the data differently. As the business grows, the headquarter is planning to implement a data warehouse system that can store and analyse data for decision-making. Specifically, the data manager at the headquarter would like to oversee the performance of different stores, and be able to analyse customers' behaviour.

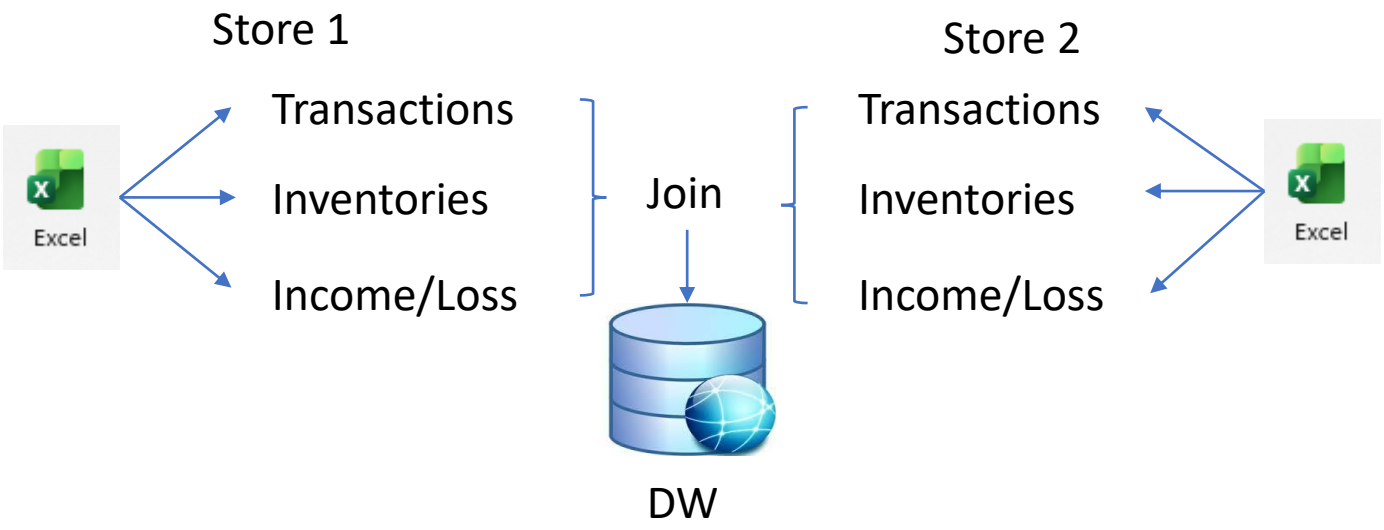


Figure 1: An overview of data integration in PhoneFix, with two stores shown

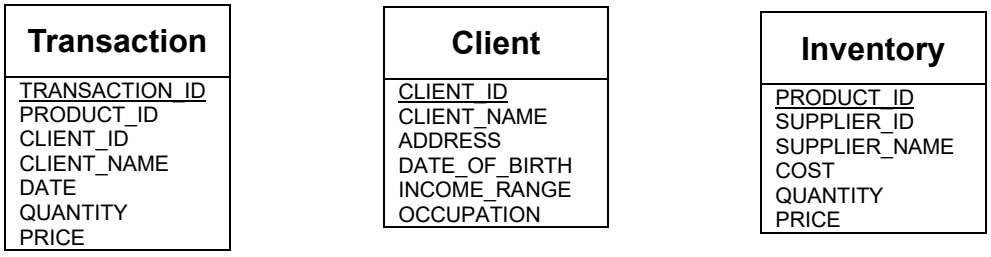


Figure 2: Three example tables used by the owner of Store 1

A.2 Data specifications

A.2.1 Current Design

Currently, the individual owners decide their own data storage method. At the end of each quarter, each owner sends an Excel file to the headquarter detailing the income/ loss, transaction records, as well as inventories. These Excel files are then joined to form a data warehouse (DW). **Figures 1** shows a schematic workflow (of course, there are more stores in reality). Some example tables used by one of the store managers is shown in **Figure 2**, where the primary keys are underlined.

A.2.2 ETL

Extraction Transformation & Load (ETL): ETL could be identified as a potential issue. However, throughout this assessment, you can assume that ETL can essentially be managed.

A.3 Schema Design

Because the IT people at **PhoneFix** have never taken COMP810, they do not know much about data warehousing. They have approached you for some suggestions to improve the current DW by designing a better schema. By considering their requirements, you are required to design a ***logical schema*** for the Data Warehouse (DW). You can choose either the STAR schema or the SNOWFLAKE schema.

Explain the deficiencies of the current DW design used by PhoneFix, justify your choice of schema and the composition of the fact and dimension tables, and explain the comparative advantages of your proposed design over the existing one.

A.3.1 Things to consider

The logical schema for the DW represents aggregated data for specific business activities. Using the schema, one can create multiple aggregated data sources that will represent different aspects of business operations. For example, the aggregation may involve total sales by selected time periods, by product, by client, and so on.

Some aggregated totals to consider are: total products sold, total sales values by product, etc. The basic star and snowflake schemas have three main components: *facts, dimensions, attributes.* Consider what constitutes the fact and dimension tables. To determine the correct dimension tables and attributes, you will need to consider the attributes shown in Figure 2. However, you do not need to limit yourself to the attributes shown. Feel free to use your imagination and creativity to make your schema more suitable for the client.

A.4 Drawing DW Schema

Use a Data Modelling software to draw the DW Schema, such as Microsoft Visio or EDRplus. Visio is available in AUT computers. If you use some other Data Modelling software, be aware that all required elements of the ERD may not be clearly shown.

A.5 Creating the Tables

Hence, create the required tables using Oracle. You do not need to insert any records in the tables. Submit the code and the descriptions (using DESC) of all tables created (e.g., by screenshots).

A.6 What to submit

To summarise, for this section, you need (1) explanations and justifications of your design [this should be written concisely with not more than 2 pages excluding references, if any], (2) a diagram representing the schema for the DW, (3) the Oracle SQL code needed to create the tables, and (4) the descriptions (using DESC) of all tables created. Check the rubrics on Canvas for a detailed breakdown of the marks in this section.

Section B — Multidimensional Data Analysis (50 marks)

In this Section, we concentrate on multidimensional data analysis. We use the SH Sample Schema available in SQL (with prefix 'sh.'; e.g., you can bring the sales table up using `SELECT * FROM sh.sales`). The schema diagram is available on Canvas (you can ignore the costs table for this assessment).

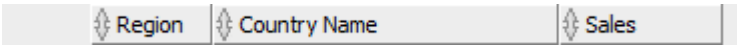
B.1 Instructions

Answer the following questions. Write appropriate SQL code to generate the results. Include both the query (SQL code), the output (e.g., screenshots) and a conclusion in your answer.

Marks: 6 marks each for Q1-Q5 (4 marks code and output + 2 marks conclusion in writing), and 10 marks each for Q6-Q7 (5 marks code and chart + 5 marks writing).

B.2 Short-answer questions

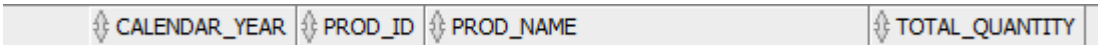
Question 1 Find and display the top five countries with the highest total sales (based on amount_sold) in 2000 and 2001. Display the regions of the countries (country_region) where the customers were based, names of these countries (country_name), and the total sales amount (rounded to the nearest dollar, presented in a format like '$123,456,789.00') as follows:



Question 2 By using GROUP BY ROLLUP, find and display (as follows) the total sales amount of all types of CD-R or CD-RW sold in all countries in Europe. In displaying the sales amount, replace the decimal places by 'xx' (i.e., in a format like '$123,456,789.xx'). Display the output as follows:



Question 3 Write a query involving a nested query to find and display the product that was sold the most (in terms of quantity) in Australia, in each year between 1998 and 2001 (inclusive). Display the output as follows:



Question 4 Find and display the top three most common sales channels (according to quantity) for all countries not in Europe that had a total sales amount greater than $500,000 in 1999. Display the output as follows:



Question 5 Create a materialised view named “Promotion_Analysis_mv” that summarises the total sales amount of all products under each promotion category in each calendar month between 1998 and 2001 (inclusive). Order by calendar_month_desc in ascending order, then total sales in descending order. The column names appear below. Take a screenshot of the table and crop it to show only the first 10 records. Briefly explain what the materialised view shows.



B.3 Long-answer questions

Question 6 Use the materialised view created in Q5 to make a chart report comparing the total sales under each promotion category. The management is deciding on which kind of promotion (if any) the company should focus on in the first quarter of 2002. Based on the information, write 3 – 4 sentences making a recommendation to the management.

Question 7 Use the materialised view created in Q5 to make a chart report summarising some information of your choice against time. The information should be significantly different from that used in Q6. Use this chart to provide some recommendations for the management. Write 3 – 4 sentences describing (a) why do you think this information would be useful, and (b) your recommendations.

Submission Requirements and Presentation (8 marks)

Submit the following to Canvas:

- A Word/Pdf file with your answers (including all necessary diagrams and charts), SQL code and output and answers for both Sections A and B.
- An SQL script named 'createDW.sql' with your statements used to create DW tables for Section A, and
- An SQL script named 'queriesDW.sql' with all your queries for Section B.

NOTES:

(a) Although it's a requirement to submit the .sql files, we will only mark the Word/ PDF file. The .sql files are for cross-checking purposes, in case needed, only.
(b) A rubric is available on Canvas.
(c) The overall structure and quality (including writing, grammar, etc.) of the report should be at an acceptable professional standard.
(d) Figures and Tables, if any, use appropriate captions and refer to them in your answers.
(e) Include references, if any, in the References section.
(f) Include an **AI Use Log** in the Appendix.
(g) You may not receive the full 8 marks if any of the above is not satisfied.

Additional Submission Information

The 'Data Warehouse Project' must be submitted to Canvas and will undergo Turnitin checks. This means that your report will be assessed through the Turnitin system, and in case of plagiarism, the University policy against plagiarism will be applied. The teaching staff reserves the right to file an 'Integrity Breach Report' if we obtain sufficient evidence confirming this.

You can register for academic integrity and paraphrasing training in the AUT Library and search for online resources. If you are using any material or figures in the assessment that is not your own, do remember to cite/reference the source.

Two weeks before the deadline, a Canvas submission link will be available for students to submit their work. You can submit your assignments as many times as you like before the deadline and you will receive a Similarity Report from Turnitin for your guidance. All content **must** be of your authorship.

Late Submissions

As per policy, failure to submit the assignment on time without special consideration approved will result in a penalty in accordance with the faculty policy (i.e., 5% per day up to a maximum of 5 days). If you experience any extenuating circumstances that prevent timely submission, please apply for special consideration via Canvas.

If you intend to lodge a Special Consideration Application, please hold your assignment submission. If the student submits an SCA along with the assignment before the due date, the late submission will NOT be graded.

Extension

Extensions will only be considered when a Special Consideration Application is made via Canvas. When in doubt, check with your instructors and the Programme Leader.

Plagiarism

- Plagiarism: If the assignment is classified as plagiarism (the overall similarity score is above 25%), all the items will be given 0. Do check your similarity report immediately after submission. Revise the report and resubmit it if it doesn't meet the requirements. If a particular section has a very high similarity score, a penalty for that section will be applied. Copying data analysis from online resources, e.g., Kaggle, will be considered plagiarism.
- Plagiarism with GenAI: If we suspect a significant portion of the report has been generated by a GenAI tool, we reserve the right to require the students to present and discuss the big data report in person through an interview format.
- Missing AI Use Log or Code: a 10% penalty will be applied if the report is submitted without the required Declaration of AI Use (AI Use Log) or code for data analysis. This applies whether or not any AI tool was used, as a declaration stating that no tool was used is still required.

Assignment Feedback

Grades for the assignment, together with feedback, will be available on Canvas upon completion of marking. This will take approximately 2 to 3 weeks.

Regulations for GAI tools

What Students Can Do with Generative AI Tools Like ChatGPT:

- **Idea Generation:** Students can use GenAI to brainstorm ideas or get suggestions to choose their report topic. This tool can be beneficial in providing different perspectives on a subject. Students are required to search Google Scholar to verify any idea provided by the GAI.

- **Research Assistance:** GenAI can provide summaries and explanations of complex concepts, which can be beneficial for students in their initial stages of research. It can also suggest further reading or resources to deepen their understanding. However, all the suggested readings MUST be verified by the student.

- **Technical Support:** Students can use GenAI to troubleshoot and seek solutions to technical problems they might encounter during their projects. By posing their issues as questions, GenAI can offer potential solutions or direct students to relevant resources.

- **Proofreading & Editing**: Students can use GenAI to check their grammar, punctuation, and spelling. It can provide suggestions for rephrasing sentences or improving the flow of the report.

What Students Should NOT Do with GenAI Tools Like ChatGPT:

- **Plagiarize**: Students are not allowed to use the output from GenAI without giving proper attribution or without adding their own understanding and analysis. Academic integrity is paramount; directly copying from an AI output is considered plagiarism. Specifically, students are prohibited from utilising GenAI to produce paragraphs that are then directly copied into their reports.

- **Sole Source of Information**: GenAI should not be the sole source of information for a report. Students should verify any facts or information from GenAI against other reliable academic sources, such as Google Scholar.

- **Over-reliance**: DO NOT over-rely on GenAI to provide ideas. The tool should be used as an assistant, not a substitute, for their own critical thinking and writing skills.

- **Unverified Information**: Students should not directly take the information provided by GenAI, e.g., generate whole paragraphs of related works. It's crucial to verify information using reputable, peer-reviewed sources, as GenAI tools can contain errors or outdated information.

Mandatory Declaration of AI Use (AI Use Log)

Every submission must include a short AI Use Log in the report appendix (the appendix is not counted towards the page limit). The log is required for the submission. For each generative AI tool used, state the tool's name, the sections in which it was used, and what it was used for (e.g., brainstorming the topic, explaining a concept, or proofreading). If you did not use any AI tool, you must still include the log and state that no AI tool was used. The log itself is not marked for content, but an honest and specific log is expected. Where the log is inconsistent with the submitted report, or where the report shows signs of unverified AI-generated content, the case may be referred under the academic integrity process, and the in-person interview described below may be required.

Suggested format for each entry: Tool (e.g., ChatGPT) | Section(s) used | Purpose | How the output was verified or revised.

For questions, please contact ryan.ip@aut.ac.nz

END OF DW PROJECT DESCRIPTION